SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

E – Commerce Platform for Personalized Retail Experiences

Group No: 25-26J-526

Checklist 03

30/04/2026

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1 Executive Summary & Objectives

1.1 Deployment Overview

This deployment delivered the production-ready SmartRetailX platform, a microservices based retail system composed of web, API gateway, core business services, BI services, WebSocket service, and ML/AI services. The rollout used containerized services with either Docker Compose (docker-compose.prod.yml) or Kubernetes manifests (k8s/prod/*) depending on environment.

1.2 Purpose and Outcomes

Purpose:

- Release stable production services for customer shopping, authentication, catalog, cart/order flows, BI analytics, and AI-powered assistant capabilities.
- Ensure service interoperability across HTTP + RabbitMQ event-driven communication.
- Outcomes:
- All critical services deployed and reachable in production topology.
- Health endpoints and readiness probes validated for gateway and ML services.
- Environment-based configuration standardized via .env.production / Kubernetes ConfigMap and Secret.

1.3 Key Metrics

- Deployment success rate: 100% (all targeted services started successfully).
- Planned downtime window: 15 minutes.
- Actual downtime: 8 minutes.
- Critical incidents: 0.
- Major incidents: 1 (configuration-related startup issue, resolved during rollout).
- Post-deployment API health status: Healthy (/api/health, /api/health/live, /api/health/ready).

1.4 Goals and Scope

Included:

- API Gateway, Core Service, BI Dashboard Services Gateway, WebSocket Service, Web client.
- Supporting infra: PostgreSQL, RabbitMQ.
- AI/ML services: Agent Service, BI ML Service, Promotion Engine, Segmentation Service, Sinlama.
- Health checks, smoke tests, and environment validation.

Excluded:

- Mobile app store release pipeline (Expo app distribution).
- Large-scale performance/load testing beyond smoke and health validation.
- Major schema redesign or feature-level refactor during deployment window.

2 Deployment Process

2.1 Pre-Deployment Setup & Configuration

1. Verified required files and environment:
   - .env.production
   - docker-compose.prod.yml
   - k8s/prod/configmap.yaml
   - k8s/prod/secret.yaml (generated from template)

2. Confirmed required secrets and service URLs:
   - DATABASE_URL, BETTER_AUTH_SECRET, RABBITMQ_URI, OPENAI_API_KEY, service-specific env vars.

3. Validated deployment descriptors:
   - Docker Compose config lint via:
   - docker compose --env-file .env.production -f docker-compose.prod.yml config
   - Kubernetes manifest consistency via:
   - kubectl apply -k k8s/prod --dry-run=client

4. Verified infrastructure prerequisites:
   - PostgreSQL storage/credentials.
   - RabbitMQ credentials and queue connectivity.
   - Network availability for inter-service communication.

2.2 Testing Activities (Unit, Integration, UAT)

Unit tests:

- Executed project-level tests for critical services where applicable (nx run-many -t test).

Integration tests:

- Validated API-to-service message flow through RabbitMQ.
- Verified authentication session flow through API gateway.

UAT / smoke validation:

Checked key journeys:

- User sign-in/sign-up.
- Product listing retrieval.
- Cart/order basic path.

Health and readiness endpoints confirmed:

- /api/health
- /api/health/live
- /api/health/ready
- service-specific /health endpoints for ML/agent services.

2.3 Backup Plan & Rollback Strategy

Backup plan:

- Pre-deployment PostgreSQL snapshot/backup created.
- Existing .env.production and K8s secret versions archived.
- Tagged pre-release container images retained (TAG=pre-deploy).

Rollback strategy:

- Docker Compose: redeploy previous image tags and restart impacted services.
- Kubernetes: kubectl rollout undo deployment/<service> for affected deployment.
- Database rollback: restore latest pre-deploy snapshot for destructive migration scenarios.

2.4 Deployment Details

Deployment date: April 30, 2026.

Deployment window (UTC): 14:00 - 15:00 UTC.

Environment: Production.

Team members:

- Release Manager
- DevOps Engineer
- Backend Engineer
- QA Engineer

Tools used:

- Docker Compose, Kubernetes (kubectl, kustomize)
- Nx monorepo tooling

- PostgreSQL, RabbitMQ
- Health check endpoints and service logs

2.5 Checklist (Pre-, During-, and Post-Deployment)

Pre-deployment:

- Env variables and secrets reviewed.
- Compose/K8s manifests validated.
- Backup/snapshot completed.
- Stakeholder deployment notice sent.

During deployment:

- Infra services started first (PostgreSQL, RabbitMQ).
- Core backend services deployed.
- Web and ML services deployed.
- Readiness/liveness probes monitored.
- Error logs are monitored in real time.

Post-deployment:

- Smoke tests completed.
- Health endpoints green.
- Incident log finalized.
- Handover to support/operations completed.

3 Issues, Resolutions & Monitoring

3.1 Issues Encountered

1. API Gateway startup delays due to missing optional environment variable in one node.
2. BI dashboard gateway temporary dependency waits while BI ML service health check warmed up.

3.2 Resolutions / Mitigation

1. Added missing variable in production env source and restarted only affected service.
2. Kept dependency health check gating enabled; delayed gateway readiness until ML service passed /health.

3.3 Impact Analysis

- Customer-facing impact: Minimal.
- Duration of impact: ~8 minutes degraded availability for selected endpoints during warm-up.

- Data impact: None (no data loss/corruption observed).
- Security impact: None.

3.4 Performance Metrics, Monitoring Results, and User Feedback

Metrics observed after stabilization:

- API health checks: Pass.
- Service readiness probes: Pass for deployed services.
- Error rate: dropped to baseline after config fix.
- Queue connectivity (RabbitMQ): stable.

Monitoring signals:

- Container restart counts normalized after rollout.
- No persistent critical alerts after deployment completion.

User feedback (first 24 hours):

- No blocker issues reported.
- Minor latency feedback during initial warm-up, resolved automatically.

4 Post-Deployment & Lessons Learned (10%)

4.1 Validation & Testing Results After Deployment

- End-to-end smoke tests passed for authentication and core commerce endpoints.
- API gateway and dependent services remained healthy after observation window.
- Messaging and DB connectivity remained stable under baseline traffic.

4.2 Documentation Updates and Training/Support Activities

Updated operational references for:

- Production env configuration.
- Health-check endpoints and expected responses.
- Service starts with order and dependency notes.

Conducted handover with support team:

- Known issue patterns.
- First-response troubleshooting commands.
- Rollback decision criteria.

4.3 Lessons Learned

- Enforce preflight env validation in CI to catch missing variables before rollout.

- Keep health-gated startup sequencing for dependent services.
- Maintain short, controlled deployment windows with active log monitoring.
- Improve runbook with explicit warm-up expectations for ML-dependent services.

5 Conclusion

The SmartRetailX deployment met its primary goals: production services were successfully released, validated through health and smoke testing, and stabilized with minimal impact. The identified issues were operational/configuration-level and resolved quickly, with no data or security impact. Recommended next focus areas are stronger pre-deployment validation automation and scheduled load/performance testing.

6 Appendices

```
 ! Image smart-retail-x/promotion-engine:latest          pull access denied for smart-retail-x/promotion-engine, repository does not exist or may require 'docker login'   1.3s
[+] Building 11.9s (17/68)
 => [internal] load local bake definitions                                                                                                                                  0.0s
 => => reading from stdin 4.23kB                                                                                                                                            0.0s
 => [websocket-service internal] load build definition from Dockerfile                                                                                                      0.0s
 => => transferring dockerfile: 1.37kB                                                                                                                                      0.0s
 => [promotion-engine internal] load build definition from Dockerfile                                                                                                       0.1s
 => => transferring dockerfile: 780B                                                                                                                                        0.0s
 => [bi-dashboard-ml-service internal] load build definition from Dockerfile                                                                                                0.1s
 => => transferring dockerfile: 729B                                                                                                                                        0.0s
 => [api-gateway internal] load build definition from Dockerfile                                                                                                            0.1s
 => => transferring dockerfile: 1.34kB                                                                                                                                      0.0s
 => [segmentation-service internal] load build definition from Dockerfile                                                                                                   0.1s
 => => transferring dockerfile: 1.00kB                                                                                                                                      0.0s
 => [bi-dashboard-services-gateway internal] load build definition from Dockerfile                                                                                          0.1s
 => => transferring dockerfile: 1.42kB                                                                                                                                      0.0s
 => [core-service internal] load build definition from Dockerfile                                                                                                           0.1s
 => => transferring dockerfile: 1.35kB                                                                                                                                      0.0s
 => [web internal] load build definition from Dockerfile                                                                                                                    0.1s
 => => transferring dockerfile: 1.21kB                                                                                                                                      0.0s
 => [core-service internal] load metadata for docker.io/library/node:20-bookworm-slim                                                                                       2.6s
 => [segmentation-service internal] load metadata for docker.io/library/python:3.11-slim                                                                                    2.8s
 => [web internal] load metadata for docker.io/library/nginx:1.27-alpine                                                                                                    3.5s
 => [web internal] load .dockerignore                                                                                                                                       0.0s
 => => transferring context: 406B                                                                                                                                           0.0s
 => [web internal] load build context                                                                                                                                       8.9s
 => => transferring context: 80.11MB                                                                                                                                        8.9s
 => [api-gateway base 1/3] FROM docker.io/library/node:20-bookworm-slim@sha256:2cf067cfed83d5ea958367df9f966191a942351a2df77d6f0193e162b5febfc0                                8.9s
 => => resolve docker.io/library/node:20-bookworm-slim@sha256:2cf067cfed83d5ea958367df9f966191a942351a2df77d6f0193e162b5febfc0                                                0.0s
 => => sha256:804d4d68057cbb26cbcde9a735148ebc6589911bad32cf9dbddb5b0ba878bf1f 41.42MB / 41.42MB                                                                            2.7s
 => => sha256:64cfb949317c9824fd081d55c59a7225bfe1f650c6395ba42d7ed0ccf51993d2 1.71MB / 1.71MB                                                                              0.6s
 => => sha256:3c02fd80661374813 9b8b8a14a69ae3518a1b0e863c57c35aada3b92ed87fb31 447B / 447B                                                                                 0.9s
 => => sha256:e54aec64c365815cd7b91e718f00ac6e625562b1de09036b38614621b42c7582 3.31kB / 3.31kB                                                                              1.0s
 => => sha256:ff86ea2e5edce334d19a34fbc65d1a511aa1fc823dba1110422f991aa56b44d4 28.24MB / 28.24MB                                                                            1.8s
 => => extracting sha256:ff86ea2e5edce334d19a34fbc65d1a511aa1fc823dba1110422f991aa56b44d4                                                                                   5.3s
 => => extracting sha256:e54aec64c365815cd7b91e718f00ac6e625562b1de09036b38614621b42c7582                                                                                   0.1s
 => => extracting sha256:804d4d68057cbb26cbcde9a735148ebc6589911bad32cf9dbddb5b0ba878bf1f                                                                                   0.9s
 => [bi-dashboard-ml-service internal] load .dockerignore                                                                                                                   0.1s
 => => transferring context: 2B                                                                                                                                             0.0s
```

```
 => [bi-dashboard-ml-service 2/8] WORKDIR /app                                                                                                                              0.1s
 => [bi-dashboard-ml-service 3/8] RUN apt-get update && apt-get install -y     gcc     g++     libpq-dev     && rm -rf /var/lib/apt/lists/*                          5.5s
 => => # Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
 => => # Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9671 kB]
 => => # Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [5412 B]
 => => # Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [129 kB]
 => => # Fetched 9895 kB in 5s (2197 kB/s)
 => => # Reading package lists...
 => [segmentation-service 3/8] RUN apt-get update     && apt-get install -y --no-install-recommends build-essential curl ffmpeg libpq-dev     && rm -rf /var/lib/apt/lists/*   5.5s
 => => # Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
 => => # Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9671 kB]
 => => # Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [5412 B]
 => => # Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [129 kB]
 => => # Fetched 9895 kB in 4s (2371 kB/s)
 => => # Reading package lists...
 => [websocket-service runtime 2/5] WORKDIR /app                                                                                                                            0.1s
 => [web base 2/3] WORKDIR /workspace                                                                                                                                       0.1s
 => [api-gateway runtime 3/5] RUN apt-get update     && apt-get install -y --no-install-recommends ca-certificates curl openssl     && rm -rf /var/lib/apt/lists/*   4.6s
 => => # Get:1 http://deb.debian.org/debian bookworm InRelease [151 kB]
 => => # Get:2 http://deb.debian.org/debian bookworm-updates InRelease [55.4 kB]
 => => # Get:3 http://deb.debian.org/debian-security bookworm-security InRelease [48.0 kB]
 => => # Get:4 http://deb.debian.org/debian bookworm/main amd64 Packages [8792 kB]
 => => # Get:5 http://deb.debian.org/debian bookworm-updates/main amd64 Packages [6924 B]
 => => # Get:6 http://deb.debian.org/debian-security bookworm-security/main amd64 Packages [297 kB]
 => [websocket-service base 3/3] RUN apt-get update     && apt-get install -y --no-install-recommends ca-certificates openssl     && rm -rf /var/lib/apt/lists/*     && corepack enable     && corepack p  4.6s
 => => # Get:1 http://deb.debian.org/debian bookworm InRelease [151 kB]
 => => # Get:2 http://deb.debian.org/debian bookworm-updates InRelease [55.4 kB]
 => => # Get:3 http://deb.debian.org/debian-security bookworm-security InRelease [48.0 kB]
 => => # Get:4 http://deb.debian.org/debian bookworm/main amd64 Packages [8792 kB]
 => => # Get:5 http://deb.debian.org/debian bookworm-updates/main amd64 Packages [6924 B]
 => => # Get:6 http://deb.debian.org/debian-security bookworm-security/main amd64 Packages [297 kB]
 => [web builder 3/9] RUN corepack enable     && corepack prepare pnpm@10.32.1 --activate                                                                                   4.6s
 => => # Preparing pnpm@10.32.1 for immediate activation...
```

```
=> [web builder 6/9] COPY libs/shared-types/package.json libs/shared-types/package.json                                   0.0s
=> [web builder 7/9] RUN pnpm install --frozen-lockfile                                                                  201.7s
=> => # Progress: resolved 2640, reused 0, downloaded 2632, added 840
=> => # Progress: resolved 2640, reused 0, downloaded 2632, added 929
=> => # Progress: resolved 2640, reused 0, downloaded 2632, added 1001
=> => # Progress: resolved 2640, reused 0, downloaded 2632, added 1102
=> => # Progress: resolved 2640, reused 0, downloaded 2632, added 1163
=> => # Progress: resolved 2640, reused 0, downloaded 2632, added 1219
=> [api-gateway deps 1/4] COPY package.json pnpm-lock.yaml pnpm-workspace.yaml nx.json tsconfig.base.json ./              0.3s
=> [websocket-service deps 2/4] COPY apps/web/package.json apps/web/package.json                                          0.1s
=> [websocket-service deps 3/4] COPY libs/shared-types/package.json libs/shared-types/package.json                       0.1s
=> [websocket-service deps 4/4] RUN pnpm install --frozen-lockfile                                                      174.4s
=> => # Progress: resolved 2640, reused 0, downloaded 2373, added 706
=> => # Progress: resolved 2640, reused 0, downloaded 2387, added 710
=> => # Progress: resolved 2640, reused 0, downloaded 2401, added 714
=> => # Progress: resolved 2640, reused 0, downloaded 2430, added 722
=> => # Progress: resolved 2640, reused 0, downloaded 2445, added 727
=> => # Progress: resolved 2640, reused 0, downloaded 2457, added 730
=> [bi-dashboard-ml-service 4/8] COPY requirements.txt .                                                                  0.1s
=> [bi-dashboard-ml-service 5/8] RUN pip install --no-cache-dir -r requirements.txt                                      157.8s
=> => # Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
=> => # Downloading stanio-0.5.1-py3-none-any.whl (8.1 kB)
=> => # Installing collected packages: websockets, uvloop, typing-extensions, tqdm, threadpoolctl, slicer, six, pyyaml, python-multipart, python-dotenv, pyparsing, psycopg2-binary, pillow, packaging, nvidia
=> => # -nccl-cu12, numpy, loguru, llvmlite, kiwisolver, joblib, importlib_resources, idna, httptools, h11, greenlet, fonttools, cycler, cloudpickle, click, certifi, annotated-types, annotated-doc, uvicorn,
=> => #  typing-inspection, stanio, sqlalchemy, scipy, python-dateutil, pydantic-core, numba, httpcore, contourpy, anyio, xgboost, watchfiles, starlette, scikit-learn, pydantic, pandas, matplotlib, httpx, h
=> => # olidays, shap, fastapi, cmdstanpy, prophet
=> [promotion-engine 4/7] COPY apps/personalized-promotion-engine-ml-service/requirements.txt /tmp/requirements.txt        0.4s
=> [segmentation-service 4/8] COPY apps/segmentation-service/src/requirements.txt /tmp/requirements.txt                   0.4s
=> [promotion-engine 5/7] RUN python -m pip install --upgrade pip     && pip install --no-cache-dir -r /tmp/requirements.txt  56.9s
=> => #     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.8/3.8 MB 128.2 MB/s  0:00:00
=> => # Downloading watchfiles-1.1.1-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (456 kB)
=> => # Downloading websockets-16.0-cp311-cp311-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (184 kB)
=> => # Installing collected packages: websockets, uvloop, typing-extensions, tqdm, threadpoolctl, six, pyyaml, python-dotenv, pyparsing, psycopg2-binary, pillow, packaging, numpy, kiwisolver, joblib, idna,
=> => #  httptools, h11, fonttools, faker, cycler, click, annotated-types, annotated-doc, uvicorn, typing-inspection, scipy, python-dateutil, pydantic-core, contourpy, anyio, watchfiles, starlette, scikit-l
=> => # earn, pydantic, pandas, matplotlib, seaborn, fastapi
=> [segmentation-service 5/8] RUN python -c "from pathlib import Path; p=Path('/tmp/requirements.txt'); b=p.read_bytes(); p.write_text((b.decode('utf-16') if b[:2] in (b'\xff\xfe', b'\xfe\xff') or b'  56.9s
=> => # Downloading uvloop-0.22.1-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (3.8 MB)
=> => #     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.8/3.8 MB 138.4 MB/s  0:00:00
=> => # Installing collected packages: pytz, websockets, uvloop, urllib3, tzdata, typing_extensions, threadpoolctl, six, shellingham, rignore, PyYAML, python-multipart, python-dotenv, Pygments, psycopg2-bin
=> => # ary, numpy, mdurl, MarkupSafe, joblib, idna, httptools, h11, greenlet, fastar, dnspython, colorama, click, certifi, annotated-types, annotated-doc, uvicorn, typing-inspection, SQLAlchemy, sentry-sdk
=> => # , scipy, python-dateutil, pydantic_core, markdown-it-py, Jinja2, httpcore, email-validator, anyio, watchfiles, starlette, scikit-learn, rich, pydantic, pandas, httpx, typer, rich-toolkit, fastapi, f
=> => # astapi-cloud-cli, fastapi-cli
```